

MIS 443 – In-term Test Report

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Question 1: Database Setup

A PostgreSQL database was created and the Northwind dataset was successfully imported. A new table named students was created with three attributes: studentid (Primary Key), fullname (NOT NULL), and email (UNIQUE). Personal information was inserted into the table successfully.

Question 2: Top 5 Customers by Number of Orders

An SQL query was written using JOIN, COUNT(), GROUP BY, and ORDER BY to identify the top five customers who placed the highest number of orders. The results were sorted in descending order of total orders.

Question 3: Orders and Customers

An SQL query was created to display orders together with customer information. The customers and orders tables were joined using customer_id, and the results were sorted by order_date in descending order.

Question 4: Large Quantity Orders

An SQL query was developed to identify orders where a product was purchased in quantities greater than 99 units. The query joined orders, order_details, and products to display order information, product names, and quantities.

Question 5: Average Delivery Time by Shipper

An SQL query was written to calculate the average delivery time for each shipping company. Delivery time was calculated as the difference between shipped_date and order_date, and the average delivery days were computed using the AVG() function.

Question 6: Customer Ranking

An SQL query was created to rank customers based on the total number of orders they placed. The RANK() window function was used so that customers with the same number of orders received the same rank.

Conclusion

The Northwind database was successfully queried using SQL techniques including joins, aggregation functions, grouping, sorting, filtering, and window functions. The results provided useful insights into customer activity, order patterns, and shipping performance.

Github Link: [https://github.com/tuyetnguyenthianhbbs23-collab/MIS-443/tree/main/MIS%20443 In-term%20Test](https://github.com/tuyetnguyenthianhbbs23-collab/MIS-443/tree/main/MIS%20443%20In-term%20Test)